TEMPEST: Transformative Explorations in Multi-Physics and Engineering of Scientific Turbulence

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Partner Institutions: Michigan State University (Lead), Auburn University, Yale University, Baylor University, Georgia Institute of Technology, University of Rochester, San José State University, Texas A&M University Corpus Christi

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(3) Project Description

(3a) Problem Description and Rationale for Center Approach

Turbulence is the defining challenge of 21st-century science. It shapes weather and climate, governs energy conversion in fusion and astrophysical systems, and dictates performance in engineered systems from engines to environmental flows. For decades, progress has come through studying canonical flows—idealized geometries with single-physics dynamics and clean scale separation. These foundational studies enabled remarkable advances in simulation, modeling, and control. Yet, the flows that matter most to today's grand challenges are anything but canonical.

Non-canonical turbulence arises where multiple physical processes interact—magnetic fields, radiation, chemical reactions, kinetic effects, non-Newtonian rheology—across irregular geometries and extreme gradients. These flows govern energy confinement in tokamaks, turbulent mixing in inertial fusion, severe weather formation, and stellar explosions that synthesize universal elements. Unlike canonical flows, they are inherently multiscale and multiphysics, violating the simplifying assumptions that enabled past progress.

At the heart, lies the **triple closure problem** (Figure 1): three successive upscaling steps requiring closure modeling without clear separation. First, molecular dynamics transitions to kinetic descriptions through the Bogoliubov-Born-Green-Kirkwood-Yvon (BBGKY) hierarchy—requiring collisional closures to reduce intractable N-body systems in $6N$ dimensions to manageable phase-space models. Second, kinetic theory connects to hydrodynamics through moment closures that require assumptions about velocity distributions. Third, turbulence closures—including Reynolds-averaged Navier-Stokes (RANS) models, large-eddy simulations (LES), and reduced-order representations—bridge resolved hydrodynamic scales to unresolved turbulent dynamics that direct numerical simulations cannot capture.

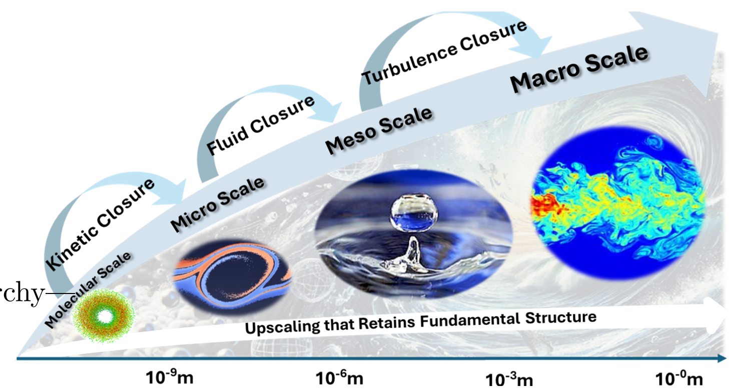

Figure 1: Multi-scale triple closure hierarchy: from particle interactions to turbulent flows via kinetic and hydrodynamic bridges.

In each transition, traditional theory imposes sampling assumptions—ergodicity, isotropy, near-equilibrium—that fail in non-canonical settings. With strong forcing, anisotropic confinement, sharp gradients, or intermittency, standard closure methods fail catastrophically. *The triple closure problem represents fundamental breakdown of conceptual tools for moving from microscopic laws to macroscopic predictions.*

TEMPEST confronts this challenge through integrated rigorous physics-preserving artificial intelligence (AI) and machine learning (ML), breakthrough theoretical frameworks, and revolutionary experimental platforms. Our goal transcends advancing individual domains—we build coherent scientific architecture for understanding, modeling, and predicting turbulence when traditional assumptions collapse.

TEMPEST unites leading capabilities across experimental science, theory, and computation. Experimental platforms at Michigan State, Rochester, Auburn, and Baylor directly access non-canonical regimes: a variable-viscosity tunnel controlling scale separation, laser-driven plasma facilities probing molecular-to-fluid transitions, dusty plasmas with tunable correlation, shock-driven compressible mixing systems, and a volumetric energy deposition chamber creating time-dependent instabilities. These co-designed facilities ensure physically meaningful, model-informing data.

Theoretical foundations include moment-based turbulence descriptions without single-physics closures, mathematical tools for nonlocal interactions and multiscale couplings, and first-principles reduced models retaining essential cross-regime structure. Crucially, proposed AI and ML provide structure-preserving algorithms that encode conservation laws, respect symmetries, preserve geometry, and integrate with physical models. This extracts closure models, surrogate dynamics, and interpretable structures from complex data without sacrificing fidelity—AI bridges theory and experiment rather than circumventing physics.

Central to TEMPEST is the assertion that **data leads transformation.** TEMPEST creates curated, high-fidelity open datasets serving the scientific community and public: experimental data from unique multi-physics testbeds and simulation data spanning kinetic-to-turbulence regimes. Structured, documented releases in reusable formats facilitate cross-comparison and benchmarking. This shared foundation catalyzes discovery beyond the center, enabling new AI models, accelerating theory, and supporting reproducible science across disciplines.

TEMPEST recognizes that problems of this scope demand collaboration across fluid dynamics, plasma physics, applied mathematics, machine learning, and experimental science. The structure fosters genuine integration through shared cohorts, standardized protocols, and co-developed tools. TEMPEST trains next-generation scientists operating fluently at physics-data-experimentation intersections. *TEMPEST's legacy:* unified framework for non-canonical turbulence transcending individual applications and establishing new multi-scale, multi-physics science standards. Impact spans accurate fusion models, extreme weather prediction, improved energy efficiency, and insights into cosmic matter origins.

Enormous strides have been made toward mastering canonical turbulence. TEMPEST takes the essential next step—confronting real-world turbulence and building integrated predictive science to understand and control it.

(3b) Research Objectives of the Center

TEMPEST's three research pillars are designed to attack the triple closure problem from complementary directions that must be tightly integrated to succeed. Our experimental pillar will generate direct observations of non-canonical turbulence phenomena across unprecedented ranges of scale separations and multi-physics interactions. Our AI/ML pillar will develop the mathematical frameworks and computational tools needed to extract predictive insights from high-dimensional,

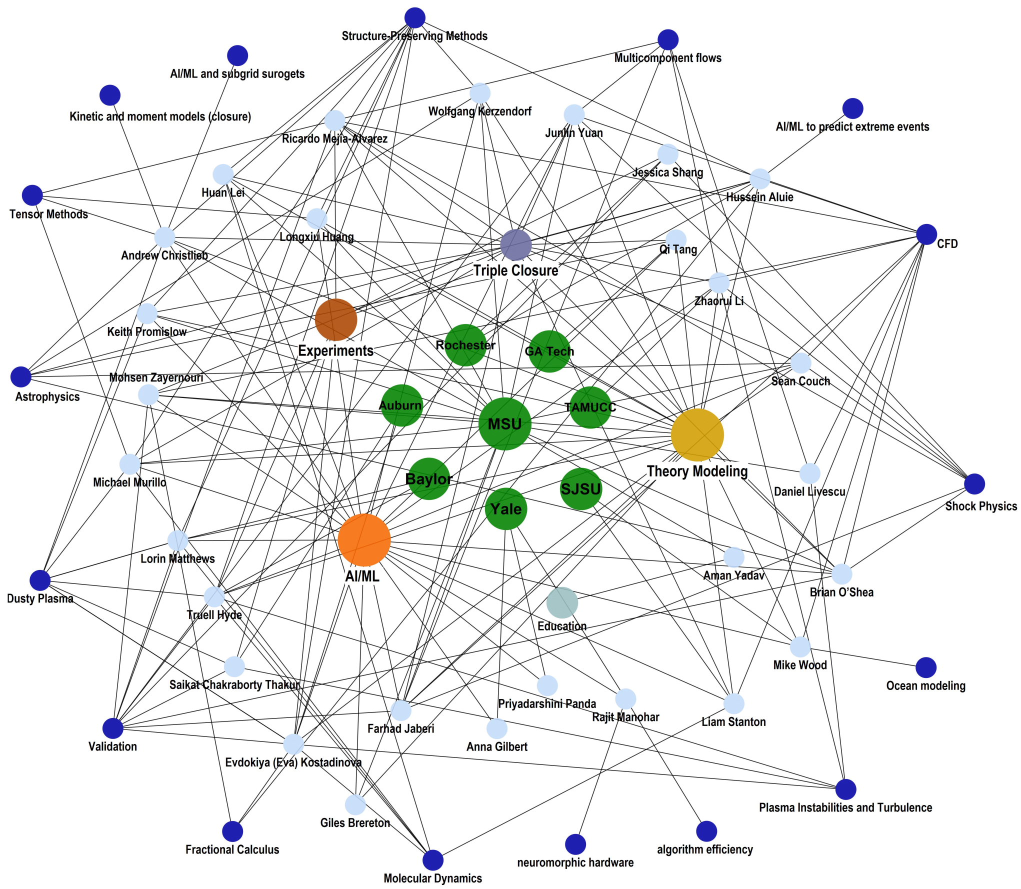

Figure 2: TEMPEST Collaborative Network: Dense interconnections between institutions, researchers, and research topics demonstrate the center's highly integrated, cross-disciplinary approach spanning multiple geographic regions and institutional boundaries.

multi-scale, multi-physics datasets. Our theory/modeling pillar will create the next generation of closure models that bridge atomic-scale interactions to continuum behavior while simultaneously capturing multiple competing physical effects.

TEMPEST's three research pillars achieve transformative integration through the dense collaborative network illustrated in Figure 2, where researchers, institutions, and research topics form interconnected nodes that span multiple geographic regions and disciplinary boundaries. This network structure ensures that experimental data from the revolutionary facilities shown in figure 3 [Dusty Strongly-coupled Turbulent Experimental Complex (DUSTEC), Advanced Blast Chamber (ABC), Boundary Layer Shock Tunnel (BLAST), Volumetric Energy Deposition (VED) facility, Adaptive Viscosity Flow Tunnel (AVFT), Linac Coherent Light Source (LCLS)] directly informs AI/ML algorithm development for structure-preserving neural networks and physics-informed closures, which in turn validate and enhance theoretical frameworks spanning molecular to fluid scales. The tight coupling visible in Figure 2 enables rapid knowledge transfer: experimental observations of non-canonical phenomena immediately feed AI/ML surrogate development, these surrogates create novel closures for theory and modeling frameworks, and enhanced models guide the design of next-generation experiments across all partner institutions. Critically, our "inreach" strategy—

embedding researchers across different pillars through month-long rotations, cross-institutional mentoring, and shared facility access—ensures that breakthrough discoveries in one domain rapidly propagate throughout the network. This integrated approach pushes the boundaries of what is possible by building smarter closures at each scale, enabling upscaled models to capture the correct emergent dynamics for non-canonical systems while fostering genuine interdisciplinary collaboration that transcends traditional academic silos. The network's density and cross-connections, evident in the visualization Figure 2, represent our commitment to creating a unified framework where experimental validation, AI/ML innovation, and theoretical advancement occur simultaneously across the full spectrum of turbulence scales.

❂ Experiment

Experimental Research Overview

Vision: Generate unprecedented multi-scale, multi-physics datasets that directly address fundamental limitations of canonical turbulence theory. **Areas:** (1) Multi-scale particle tracking to macroscopic structures; (2) Multi-physics electromagnetic-kinetic and thermal-momentum coupling; (3) Triple closure problem across kinetic, hydrodynamic, and turbulence hierarchies. **Tools:** DUSTEC, ABC, BLAST, AVFT, VED, LCLS. **Impact:** Transform turbulence validation through direct observation of non-canonical phenomena, providing empirical foundations for fusion energy, climate dynamics, and engineering applications.

TEMPEST's experimental thrust generates unprecedented multi-scale, multi-physics datasets that directly address the fundamental limitations of canonical turbulence theory by probing regimes where traditional assumptions break down. Our revolutionary experimental capabilities provide essential validation data for predictive models across the vast scale separations and complex physics interactions at the heart of the triple closure problem.

Six complementary experimental platforms (shown in Figure 3) address TEMPEST's three core challenges through coordinated investigation. For the **Multi-scale Challenge**, DUSTEC at Auburn provide high-precision particle tracking from individual kinetics to macroscopic structures, while the ABC and BLAST at Michigan State bridge microscopic dissipation to macroscopic shock structures in three-dimensional configurations. The **Multi-physics Coupling Challenge** is tackled through the AVFT investigations of thermal-momentum coupling across six orders of magnitude in viscosity ratios, the VED studies of energy-density-momentum interactions through time-dependent instabilities, and the ABC facility's capture of bidirectional shock-turbulence coupling. For the **Triple Closure Problem**, DUSTEC directly observes non-Maxwellian distributions and correlation effects violating molecular chaos assumptions, while the LCLS uses X-ray photon correlation spectroscopy to probe transport phenomena in warm dense matter, where local thermodynamic equilibrium breaks down completely.

This coordinated approach transforms turbulence science validation through revolutionary experimental capabilities that probe real-world, non-canonical, turbulent systems, providing the empirical foundation that traditional canonical flow studies cannot deliver.

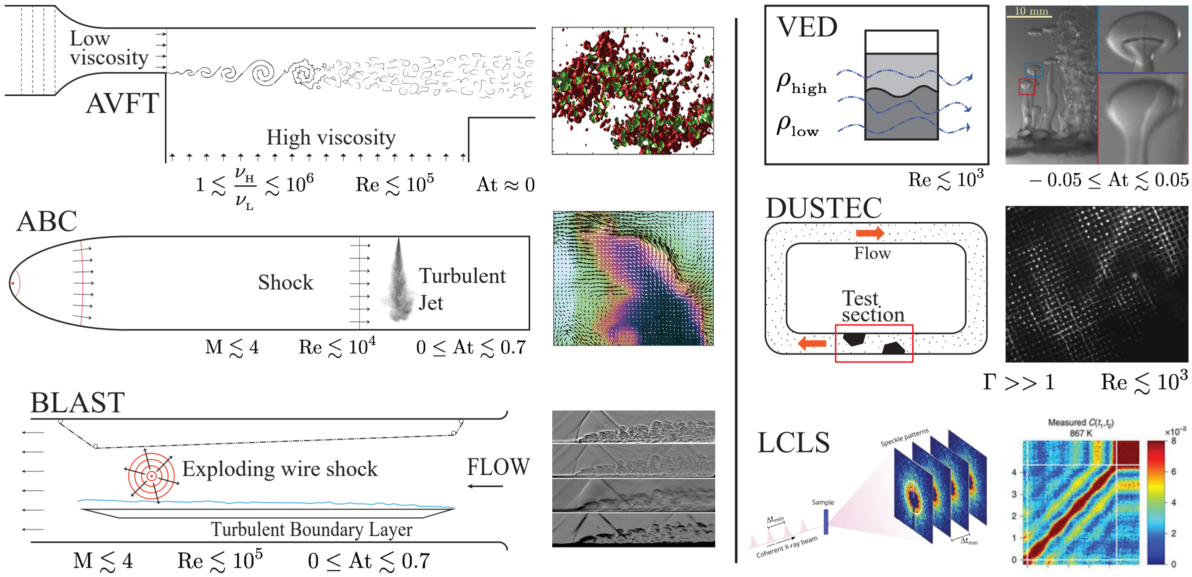

Figure 3: In addition to data from other sources, TEMPEST employs six experimental platforms: AVFT–variable viscosity turbulence, ABC & BLAST–shock-turbulence interactions, VED–turbulent mixing driven by volumetric energy deposition, DUSTEC–strongly coupled turbulent dusty plasmas, LCLS–transport phenomena in extremes.

➢ **Multi-scale Challenge:** TEMPEST's experimental thrust directly attacks the multi-scale challenge through unprecedented capabilities that enable simultaneous observation across the vast scale separations where traditional turbulence theory breaks down. Our coordinated experimental platforms offer the first systematic approach to validate theoretical models across critical intermediate scales—from individual particle dynamics in dusty plasmas to shock-driven macroscopic structures in blast chambers—where mesoscopic interactions fundamentally alter turbulent behavior but remain experimentally inaccessible by conventional approaches.

Team's Expertise: Our experimental team has established world-leading capabilities for investigating scale-coupling through the systematic development of multi-scale diagnostic techniques. Key contributions include precision particle tracking in dusty plasma systems, enabling direct observation of kinetic-to-fluid transitions [59, 36], construction of the nation's largest university-owned blast chamber for three-dimensional shock-turbulence interactions [37], and pioneering variable viscosity experimental techniques with time-resolved, three-dimensional diagnostics [18]. Our dusty plasma work has revealed cross-scale coupling effects leading to global collective behavior, including superdiffusive transport under strong magnetic fields up to 3000T [20] and helical structure formation in particle chains [21].

Proposed Work: (i) Dusty plasma experiments at Auburn will quantify anomalous diffusion effects using discrete fractional Laplacian methods, providing experimental validation of non-extensive statistical approaches and foundational data for closure models spanning kinetic-to-fluid transitions, supporting work in both the theory/modeling and AI/ML pillars. **(ii)** ABC facility studies will investigate cross-scale energy transfer by examining how microscopic dissipation couples to macroscopic shock structures through vorticity transport, exploiting genuine three-dimensional effects unavailable in idealized configurations. **(iii)** Variable viscosity experiments will investigate thermal-momentum coupling and cross-scale feedback loops, validating spectral transport approaches and examining how structural disorder alters energy and momentum transport pathways. **(iv)** X-ray photon correlation spectroscopy at LCLS will directly measure scale-dependent transport coefficients in warm dense matter, validating theoretical and AI/ML frameworks for non-local transport where local thermodynamic equilibrium fails. **(v)** Coordinated multi-facility campaigns will establish universal scaling relationships across physical systems, distinguishing fundamental scale coupling mechanisms from system-specific phenomena.

➢ **Multi-physics Coupling Challenge:** TEMPEST's experimental thrust confronts the multi-physics coupling challenge by providing the first coordinated platform to systematically isolate, control, and measure the complex interactions between competing physical effects that govern non-canonical turbulence. These multi-physics measurements will simultaneously provide our AI/ML thrust with datasets that capture essential coupling signatures for developing physics-informed neural networks and validate our theory/modeling closure frameworks.

Team's Expertise: Our experimental team has established capabilities for investigating multi-physics coupling through systematic development of simultaneous multi-field diagnostic techniques. Key contributions include groundbreaking dusty plasma work revealing how electromagnetic forces compete with fluid inertia and ion density waves drive filamentary structure formation [38, 14, 9, 60], revolutionary X-ray photon correlation spectroscopy capabilities for measuring transport coefficients in warm dense matter [63], world-leading expertise in shock-turbulence interactions through advanced blast facilities [37], and innovative development of volumetric energy deposition capabilities for studying energy-density-momentum coupling in time-dependent instabilities [45]. Our work has explored the electromagnetic-kinetic coupling in magnetized systems [27].

Proposed Work: **(i)** Dusty plasma experiments will quantify electromagnetic-kinetic coupling effects that create non-local interactions violating traditional fluid assumptions, providing validation for coupled electromagnetic-fluid closure models. **(ii)** X-ray photon correlation spectroscopy studies will measure coupled advection-diffusion-shear processes in laser-generated plasmas, expanding the pioneering experiment conducted in March 2024 to validate closure relationships connecting kinetic-scale to fluid-scale phenomena. **(iii)** Shock-turbulence interaction studies across multiple configurations will quantify compressible-turbulent coupling via pressure-dilatation, acoustic emissions, and density gradient effects. **(iv)** Magnetized and unmagnetized dusty plasma studies will establish how magnetic field topology influences particle transport and energy dissipation, validating electromagnetic-plasma coupling models. **(v)** Cross-facility campaigns investigating thermal-momentum, electromagnetic-kinetic, and energy-density coupling will distinguish universal mechanisms from system-specific phenomena, providing foundations for closure models capturing complex multi-physics interactions.

➢ **Triple Closure Challenge:** TEMPEST's experimental thrust addresses the triple closure problem by providing direct observational evidence of closure model failures across all three hierarchical levels (Figure 1) where traditional assumptions break down catastrophically in non-canonical systems. Our unique experimental capabilities provide our AI/ML thrust with essential datasets for developing structure-preserving neural networks and our theory/modeling thrust with validation benchmarks for next-generation closure frameworks.

Team's Expertise: Our experimental team has established unique capabilities for directly probing closure model failures through systematic development of techniques accessing previously inaccessible physical regimes. Key contributions include pioneering dusty plasma work, enabling direct phase space reconstruction through individual particle tracking [24, 43], revolutionary capabilities for studying self-diffusion in strongly coupled systems, revealing superdiffusive behavior and anomalous scaling [20], and world-leading expertise in extreme conditions using X-ray photon correlation spectroscopy [1, 28, 63]. Our work has advanced understanding of how structural disorder alters transport properties through spectral approaches [25, 26].

Proposed Work: **(i)** Dusty plasma experiments will characterize non-Maxwellian distribution functions and quantify correlation effects using discrete fractional Laplacian approaches, providing foundations for non-extensive statistical closures and modified kinetic theories. **(ii)** Variable viscosity and volumetric energy deposition studies will investigate transport modifications under non-equilibrium conditions, validating closure relationships that capture non-local transport effects and correlation-induced modifications. **(iii)** Shock-turbulence interaction studies will measure how coherent structures modify turbulent transport through vorticity transport and pressure-dilatation coupling, providing validation data for structure-mediated transport closure models. **(iv)** X-ray photon correlation spectroscopy will measure transport scaling relationships under conditions where molecular chaos, local equilibrium, and ergodic assumptions simultaneously break down, providing critical validation data for closure model failures. **(v)** Cross-platform integration campaigns will validate closure models across multiple physics regimes—dusty plasma correlations, extreme-condition transport, and shock-turbulence interactions—establishing universal principles while identifying regime-specific modifications.

Research Impact

Intellectual Merit: Revolutionary experimental capabilities enabling direct observation of non-canonical turbulence where traditional assumptions fail

Broader Impacts: Open-source datasets and codes advancing weather prediction, fusion energy, and astrophysical modeling

Integration Across Themes: Validation data for AI/ML neural networks, benchmarks for theory/modeling closures, training datasets for researchers

Feasibility: World-leading facilities operational with proven multi-scale diagnostics and established collaborations

Legacy: Transform turbulence validation through unprecedented multi-physics datasets spanning kinetic to continuum scales

❂ Theory/Modeling

Theory & Modeling Research Overview

Vision: Establish advanced theoretical, modeling and computational tools that bridge disparate time and length scales and multiple physics regimes for turbulence applications. **Areas:** (1) Reduced-order and nonlocal dynamics; (2) Coarse graining to bridge physics regimes and scales; (3) Advanced numerical simulations of turbulent conditions. **Tools:** Fractional calculus, coarse graining, ML hybrid solvers, BBGKY hierarchy, closure models, Mori-Zwanzig formalism, exascale computing. **Impact:** Enable novel modeling capabilities for non-canonical turbulence with extreme scale separation, advance structure-preserving ML through new designs, optimize experimental validation efforts.

The theory and modeling pillar addresses the core challenge of developing closure models that can simultaneously represent multiple competing physical effects without the scale separation assumptions that enable single-physics approaches. Our strategy combines micro-meso-macro-scale modeling to create multi-scale, multi-physics frameworks that are suitable for the full range of non-canonical flow conditions. The proposed work encompasses a host of modeling efforts under the theme of non-canonical turbulence informed by physics at all scales.

Drawing from the expertise of our team in theory and modeling across multiple physical domains, this pillar consists of 3 thrusts, each addressing a component of the multi-scale, multi-physics, and triple closure challenges. 1) Expand the constraints of both modeling and computation of turbulence applications through novel reduced-order and nonlocal dynamics; 2) Develop coarse-graining frameworks for micro-, meso- and macro-scale physics; 3) Extend hydrodynamic models, kinetic theories, and moment closures for novel models that enable new multi-scale and/or multi-physics numerical simulations, leveraging exascale computing, AI/ML hybrid algorithms, and verification and validation with experimental data. The proposed thrusts in this pillar will work synergistically with experimental facilities, shaping experimental design to optimize the value of validating data.

➢ **Reduced Order and Non-local Dynamics:** Enhance turbulence modeling through reductive techniques that identify and incorporate innovative non-local effects.

Team's Expertise: Our team has pioneered fractional calculus approaches for turbulence modeling, developing α-stable Lévy distribution models for nonlocal subgrid effects, showing that fractional Laplacian operators can capture heavy-tailed intermittent dynamics that traditional models often miss [3, 4] with applications to LES subgrid modeling, filtered Boltzmann transport equations, and nonlocal Navier-Stokes equations for anomalous transport.

Proposed Work: The high computational cost of direct turbulence simulation necessitates reduced-order modeling, which is the focus of this thrust. **(i)** We propose a novel spatio-temporal filter for the expensive $7D$ Boltzmann transport equations that enables coarse-grained, physics-informed, reduced-order modeling. This filtered framework allows the development of tailored, nonlocal collision operators, forming the basis for a new class of hydrodynamic-scale closure models. We also address the inconsistency between the Bhatnagar-Gross-Krook (BGK)-type closures and filtered turbulent dynamics by using a fractional Laplacian closure term to capture nonlocal effects. This approach enables the development of dynamic, nonlocal LES models across various kinetic regimes, along with new spectral scaling laws and cascade models for anisotropic and inhomogeneous flows. First-principles molecular dynamics (MD) simulation data will guide the tuning of fractional parameters in the new model [3, 4]. **(ii)** We propose a novel mathematical-computational-statistical framework for multi-scale (kinetics-to-continuum) subgrid modeling. Our preliminary *a priori* and *a posteriori* studies decisively reveal that subgrid models often emerge as nonlocal operators, e.g., fractional Laplacians or generally as variable-to-distributed order turbulence models, hence outperforming the established local models in resolving the forward cascade and backscattering of turbulence and prediction of subgrid probability density functions [64]. Subsequently, we will develop high-order and hybrid splitting (Eulerian-Lagrangian-ML) schemes in the context of nonlocal Navier-Stokes equations. **(iii)** We propose new reduced models that generalize the shallow water equations and capture vertical information missing in current hurricane simulations. To address inaccuracy in the existing models under non-split bottom boundary conditions, we will extend moment expansion models suitable for both non-slip and slip boundaries. The primary difference in the new model lies in the treatment of the source term, which allows our modified moment expansion models to be readily generalized while guaranteeing strong hyperbolicity, a critical condition for well-posedness. This effort will partner with the AI/ML pillar to capture turbulent effects. **(iv)** We propose to develop a new continuum model and computational capability that incorporates spatio-temporal correlations through the use of density functional theories and fluctuating hydrodynamics [52, 42], where the necessary correlation functions and memory kernels will be trained on MD simulations and dusty plasma experiments from the TEMPEST team. This capability will serve to better understand plasmas relevant to inertial confinement fusion experiments, where collective behavior has important effects are non-negligible.

➢ **Coarse Graining to Bridge Physics Regimes and Scales:** This thrust addresses how structures couple to each other in turbulent flows, both in space-time and across scales.

Team's Expertise: We have developed the open-source coarse-graining framework FlowSieve [53] that implements a series of novel coarse-graining methodologies. We also led the development of several coarse-graining frameworks [5, 6].

Proposed Work: The proposed work will leverage the open-source coarse-graining frameworks led by our team and adapt them for turbulent flow applications. **(i)** We will implement a coarse-graining framework [65] to understand the coupled structures in turbulent flows. The novelty lies in its capability of revealing the transfer of energy between different scales in flows, which is important for inertial confinement fusion, supernova explosions, and planetary flows. Equations governing the dynamics of different scales will be derived systematically, allowing for a direct analysis of processes at those scales both analytically and also using data from simulations, laboratory experiments, and observations of natural systems. **(ii)** We will use FlowSieve to understand how planetary-scale Earth system variability is coupled to weather, including extreme events. We will use this novel coarse-graining methodology for generating geographic maps of dynamical processes at different spatio-temporal scales of oceanic and atmospheric flows. Our initial focus will be the North Atlantic Oscillation (NAO), a dominant mode of regional variability in the North Atlantic region [58]. In addition to high-resolution general circulation models, we will use the 30+ year satellite altimetric record of global ocean currents along with the fifth generation EMCWF global atmospheric dataset.

➢ **Advanced Computational Models** This thrust utilizes advanced computational techniques, with acceleration from AI/ML, to achieve high-fidelity turbulence modeling.

Team's Expertise: Our team brings expertise in shock-turbulence interactions [47], variable-property flows [54], and supersonic surface roughness studies [2]. Our team led the development of several MD models, such as the N-body code for the Dynamic Response of Ions and Dust (DRIAD) [35, 56], with ML techniques to learn the ion-wake modified dust interaction force [56, 38].

Proposed Work: A series of modeling and computing capabilities are proposed for dusty plasma, inertial confinement fusion (ICF), hypersonic flows, and astrophysics. **(i)** As benchmarks for the onset of correlations in particle systems, we propose simple models that interpolate between pair-wise mean field scalings and many-particle Boltzmann-Grad scalings. System energy is a weighted average of a pair-wise energy over a fixed neighbor-distance. Preliminary work shows that gradient flows of the neighbor-averaged energy lead to a proliferation of two-scale equilibrium [22], with fine-scale multi-particle clusters on a quasi-regular lattice of characteristic size. The clustering signals the onset of correlated motion, whose prediction will serve as a benchmark for AI/ML reductions. The analytical truncations of BBGKY hierarchies show that interparticle correlations can be captured rigorously at a perturbative level for certain classes of molecular models [17]. In addition, the formation of a lattice mirrors the crystalline structure of highly-charged dust particles, which will be carefully explored in this collaborative project through both experiments [13] and numerical simulations [24, 57]. **(ii)** We will investigate the fundamental physical mechanisms guiding the onset of turbulence in dusty plasmas at low-to-medium Reynolds numbers. The viscosity of the dusty plasma fluid, determined from nonequilibrium and equilibrium methods [8], depends on the interaction between dust grains. To address this, we propose to use our DRIAD code to model the ion wakes of the flowing dust grains in the dusty plasma experiment [35]. A ML algorithm will be applied to learn the wake-mediated interaction potential as a function of plasma density and power [38, 19]. The derived interaction potential will be used in a one-component dust model to capture the spatial and temporal scales relevant to the experiment. To address the limitation of the quasi-2D ground-based experiments, numerical simulations will be used to extend results to fully three-dimensional dust clouds. **(iii)** We propose to conduct a series of high-fidelity 2D and 3D magnetized Rayleigh-Taylor instability (MRTI) simulations, under ICF deceleration stage conditions, using a high-order two-fluid plasma solver. Due to extremely large variations in the molecular transport properties and scales in this problem, unresolved "LES" methods are necessary. For this, we will develop a novel sugbrid model based on the extension of our non-plasma multi-phase LES-filtered density function (FDF) methodology [30]. LES-FDF uses a multi-particle Lagrangian numerical method which has the capability of implementing micro-scale effects locally with reliability and accuracy not possible to achieve in grid-based Eulerian methods. **(iv)** We propose to develop RAMNS (Reynolds-averaged Maxwell-Navier-Stokes) models addressing Reynolds-averaged models of turbulent MHD flows. Initial emphasis will be on small to moderate magnetic Reynolds numbers, typical of industrial molten-metal applications. We will simulate homogeneous flows in which combinations of fluid deformations (shear, plane strain, etc.) and uniform magnetic fields are imposed impulsively on decaying turbulence and allowed to develop to a steady state. **(v)** We propose advanced modeling and computations for stellar turbulence. Turbulence in explosive nucleosynthesis remains poorly understood, and in multi-dimensional core-collapse supernovae simulations, it remains under-resolved due to the lack of suitable LES models. To this end, we propose to perform advanced direct numerical simulations (DNS), implicit LES, and LES simulations of supernova shock-turbulence interactions. This will be further advanced through the development of LES models for (M)HD simulations. This effort will be closely related to the experimental pillar with comparison against experimental data such as GW 170817 - Neutron star mergers, lanthanide opacities from LANL, IceCube neutrino observatory, and gravitational waves from LIGO. **(vi)** We will perform multi-level DNS, LES, and RANS simulations to study the shock-turbulence interactions (STI) and variable viscosity turbulence (VVT) across multiple scales, developing nonequilibrium LES/RANS models with algebraic Reynolds-stress closures [10] and nonlocal fractional Laplacian operators to capture heavy-tail, intermittent dynamics at subgrid levels [64], directly validated against our Boundary Layer Shock Tunnel and Variable Viscosity Flow Tunnel facilities. STI and VVT are critical for ICF, hypersonics, and astrophysics.

Research Impact

Intellectual Merit: Advanced computational models and theoretical frameworks enabling predictive modeling for non-canonical turbulence across extreme scale separations

Broader Impacts: Open-source codes and novel closure models advancing weather prediction, inertial confinement fusion, and astrophysics simulations

Integration Across Themes: ML-enhanced hybrid solvers for AI/ML pillar, experimental validation through ABC/BLAST facilities, educational training in advanced computational methods

Feasibility: Proven expertise in shock-turbulence interactions, plasma modeling, and established open-source development capabilities

Legacy: Transform computational turbulence through structure-preserving algorithms, multi-scale closures, and ML-enhanced hybrid modeling frameworks

❂ AI/ML

AI/ML & Data Science Research Overview

Vision: Radically transform our understanding of complex physical emergent phenomena arising from the triple closure problem via novel AI/ML frameworks. **Areas:** (1) Prediction of complex phenomena in emergent systems; (2) AI/ML inference for extreme events & control/discovery; (3) Hardware and algorithm co-design. **Tools:** Structure-preserving neural networks, low rank tensor methods, sketching, slow manifold learning, reduced-order modeling, agent enhanced AI. **Impact:** Foundational ML methods for complex physics, enabling coarse-grained models to capture sub-scale dynamics from atomic to turbulence scales while bridging traditional modeling, numerical methods, and ML/AI at all levels.

The goal of the AI/ML pillar of the center is to develop transformative AI/ML methods to predict complex physical phenomena. Novel AI/ML tools offer the promise of addressing the key challenge of enabling predictive models that capture disparate scales in non-equilibrium settings, which are at the heart of the triple closure problem.

Based on our team's leadership in AI/ML across STEM, discussed below, there are three key thrusts we pursue to realize a framework for addressing the triple closure problem. They are: 1) Prediction of complex phenomena in emergent systems, directly addressing the triple closure problem through structure-preserving surrogates that enforce key mathematical structure; 2) AI/ML inference for extreme events and control/discovery, enabling predictive science through novel sampling-based algorithms and AI agent-based models; and 3) Hardware and algorithm co-design, enabling new scientific discovery through next generation AI.

The AI/ML development requires tight synergy between highest quality experiments, modeling, and algorithm development. ML will be the driving computational tool. With these tools, we aim to transform how we predict non-canonical flows.

➢ **Prediction of Complex Phenomena in Emergent Systems:** This work extends the team's mathematically well-posed AI/ML formulations and builds on structure-preserving neural networks (NNs), transfer learning via optimal transport, low-rank tensor approximations and tensor sketching, learning closures and errors in reduced-order methods, including AI/ML enhanced Mori-Zwanzig (MZ) and fractional calculus models consistent with modeling hierarchies.

Team's Expertise: Over the past decade, our team has led the development of NN closure models that are accurate, efficient, and mathematically well-posed. When designed to preserve structure, these models capture non-equilibrium, non-perturbative dynamics far beyond training regimes. Key contributions include: symplectic NNs for long-time particle emulation [11]; Lipschitz and stabilized neural ODEs for multi-scale dynamics [48]; energy-conserving and entropy-consistent surrogates for many-body collisions [66] and MZ-based upscaling from atomic to continuum scales [33, 31]. We also pioneered hyperbolicity-preserving closures that retain kinetic structure far outside the training region [16] and hybrid turbulence models using deep learning and fractional calculus [4]. We address high-dimensionality via scalable tensor methods [67] and lead in low-rank learning [12].

Proposed Work: Key new work is proposed in structure-preserving ML (SPML). We will develop a physics-informed ML framework for reduced-order modeling that directly bridges microscale descriptions with macroscale transport. We will develop the following related threads: **(i)** In collaboration with the MSU BLAST experiments, we propose to extend the coordinate discovery in [48] to a general Fenichel normal form to study boundary layers in fluid mechanics. The proposed approach establishes an explicit fast and slow split of the dynamics through a transformation learned from data or equations that ensure long time stability. **(ii)** For modeling long time dynamics in Auburn's dusty plasma experiment, we propose to discover a surrogate for a singularly perturbed Hamiltonian system. We will leverage a symplectic coordinate transformation using the symplectic HenonNet we developed in [11]. As seen in [7], the proposed approach shall overcome the temporal scale separation in the stiff dynamics using symplectic exponential integrators. **(iii)** We propose a MZ-based ML approach that builds on our previous work [31, 33], which is in direct support of non-canonical turbulence through targeting Auburn dusty plasma experiments with correlation, the MSU AVFT and VED experiments and LCLS. A two-step ML approach involves building a meso- and macro-scale model. A new meso-scale model is proposed, describing coarse-grained MD with memory that preserves heterogeneous many-body forces and physical symmetries [33]. Next, a tensor-structured encoder with SPML for macro-scale dynamics is trained from the meso-scale model [12, 15]. This semi-supervised framework enables analytically grounded, interpretable, and efficient multi-scale modeling, demonstrated for non-Newtonian hydrodynamics [29]. **(iv)** Targeting turbulence at scale in complex flows, interstellar media, Auburn dusty plasma experiments, the MSU ABC shock chamber with variable density experiments and the MSU BLAST experiments for rough surfaces, we propose to develop blended computing, the synergy of SPML with traditional numerical models. This effort builds on our initial Fractional Calculus DEEP learning work that scales up kinetic theory to turbulence scales [4, 3] for single fluid gas dynamics. Here, we seek to include multi-physics effects including: long range fields, multi-species, Coulomb collisions, chemistry, etc. This involves developing new multi-resolution ML that is consistent with the underlying physics, including symmetry, invariance [16], eigenstructure for the coupled model [16], and correct entropy [66]. **(v)** ML-enhanced reduced models target simulation at scale in a simplified format. This extends the blended computing paradigm to augmentation of $1D$ supernova and $2D$ extended shallow water models, where models are designed to connect to ML through closures and they are enhanced by bringing $3D$ turbulence transport effects into $1D$ and $2D$. Here the ML and modeling need to be designed carefully, as the information in the volume-averaged $3D$ ML turbulence surrogate must be represented in the $1D$ and $2D$ reduced models self-consistently. **(vi)** We propose quantum-informed ML for multi-scale material modeling, using Kohn-Sham density functional theory (KS-DFT) codes (e.g., Quantum Espresso) to generate high-fidelity training data for ML-based surrogates [50]. These ML models bridge quantum-level accuracy with MD-accessible scales. These quantum-informed surrogates provide essential material property inputs—equations of state, transport coefficients, phase behavior—that feed directly into the meso- and macro-scale models described in **(iii)**, while the force-based ML method enables direct comparison with Auburn's dusty plasma particle tracking data. This establishes a complete quantum-to-continuum pipeline that supports multiple TEMPEST experimental platforms and provides quantum-accurate inputs for turbulence closure models [49]. **(vii)** We propose large-scale MD at dissipative scales to develop physics-informed subgrid closure models for macroscopic flows. Leveraging ML potentials from the above quantum-information ML thrust, we will develop unprecedented MD simulations spanning scales across the dissipative range, where traditional continuum assumptions break down but collective behavior emerges. These ML-based simulations capture mesoscale physics with quantum-level accuracy and support both full $3D$ turbulence and quasi-$2D$ configurations, replicating Auburn's dusty plasma experiments using force-matched potentials. These dissipative-scale MD simulations will reveal key physics of energy cascade, intermittency, and non-local interactions. The approach bridges atomistic and continuum scales, providing direct validation for closures developed in this thrust, while enabling seamless integration with experimental data. The resulting closures will incorporate genuine dissipative-scale dynamics beyond phenomenological assumptions.

➢ **AI/ML inference for extreme events & control/discovery:** We aim to use sparse feature identification, or "fingerprint detection", to uncover early indicators of rare events—critical, for example, in extreme weather prediction. By identifying such fingerprints, even short-term anticipation of these events becomes possible. To do this, we will leverage advanced sampling algorithms that enable agent-informed control of data collection. These methods efficiently identify energetic frequencies or modes from minimal data, using algorithms that naturally align with ML architectures—layered multiplications with nonlinear activations. This framework also guides our agent-based AI sampling strategy to generate subgrid models for complex applications, including equations of state and opacities relevant to LCLS facility and MSU's AVFT experiments.

Team's Expertise: Our team has been leading the development of informed sampling methods at scale for more than two decades [55, 40]. Furthermore, we have been developing collections of novel agent-based approaches for guiding AI informed automation of data collection. These two efforts play a critical role in guiding the center in novel discoveries and address issues at scale.

Proposed work: In support of creating new and effective AI training, we propose: **(i)** Building on our expertise in identification of sparse signals, we propose to develop a new strategy for neural network identification of rare events by creating a library of fingerprints using sparse signal identification on surface temperature data provided by ECMWF to train the neural networks for event detection. The goal is to predict an extreme cold event that brought temperatures 20–35°F below average across large portions of the United States and led to numerous record lows. We will combine this with structure-preserving NNs, in the first thrust of this pillar (Complex Phenomena in Emergent Systems), providing the NN with physical guidance on what the system is constrained by. In the first two years of this project, we will focus on the NAO, which is the dominant mode of regional variability in the North Atlantic region, influencing weather patterns over North America, Europe, and parts of the Arctic. This work will be transitioned to bring rare event detection into guiding agent informed learning in the next subsection. **(ii)** Team members are pioneering reinforcement learning (RL) agents that autonomously control MD simulations to generate comprehensive, self-consistent material property datasets. The RL agent learns to optimize MD code execution—ensuring accuracy and convergence before launching subsequent runs—enabling automated generation of thousands of simulations that systematically map interdiffusion coefficients, viscosity, and equations of state across parameter space. These AI-generated datasets provide essential input data for TEMPEST's multi-scale modeling framework, with predictions validated against the Stanton-Murillo transport model [51] and experimental measurements at the Linac Coherent Light Source. We propose to extend this autonomous simulation framework to support the MSU AVFT experiment and other TEMPEST platforms, creating a closed-loop system, where AI identifies critical parameter regions, directs MD simulations, and feeds validated transport properties into upscaled turbulence models. This work directly integrates with the first AI/ML thrust on MZ and fractional calculus modeling, leveraging intelligent data identification strategies to bridge atomistic simulations with continuum descriptions.

➢ **Hardware and algorithm co-design:** The scaling of conventional computers (CPUs) has dramatically slowed down since 2005, with single-thread performance plateauing. Today's hardware is power limited, with GPU clusters in data centers moving to expensive liquid cooling technologies in a valiant effort to keep up with computational demands. However, it is clear that TEMPEST computing needs will outstrip the pace of conventional hardware. Looking forward, the TEMPEST team seeks a paradigm shift in AI/ML via co-design that leverages recent innovations in neuromorphic computing and builds from team strengths as a path to fill this gap.

Team's Expertise: Members of the team have 20+ years experience developing co-design with expertise in asynchronous "very large scale integration" design [34], architecture [39], and neuromorphic computing [46, 44]. This has deep connections with concurrency, formal methods, programming language semantics, information theory, and cognitive systems (AI/ML).

Proposed work: In support of TEMPEST's objectives, we plan to tackle this challenge with two complementary approaches: **(i)** To reduce power costs, our team members lead the development of physics-informed neural networks (PINNs) in the context of co-design with neuromorphic computing [23]. We propose to extend this work to develop subgrid models for turbulent flows, building on the work in the first AI/ML thrust (Complex Phenomena in Emergent Systems). Developing more rigorous SPML PINNs work will follow the ideas in [41] blended with [7] and will target the MZ formalism [31, 33]. This work is tied to the next deliverable as well. **(ii)** Spike-based hardware naturally supports multi-timescale operation that can be leveraged in this project. Our prior work has produced scalable neuromorphic platforms [39], all leveraging asynchronous logic for continuous-time operation and energy-adaptive computation. We will extend this foundation by combining asynchronous logic with techniques like in-memory computing to accelerate PINN algorithms. We aim to co-design hardware that directly exploits the spatio-temporal scale diversity in the experiment data. Since computational effort varies widely per modeling timestep, asynchronous architectures are ideally suited to dynamically adapt, offering an efficient substrate for this workload class.

Research Impact

Intellectual Merit: new structure-preserving AI/ML that solve the triple closure problem by bridging atomic to turbulence scales while preserving fundamental physics

Broader Impacts: open-source ML algorithms enabling breakthrough predictions in weather forecasting, fusion energy, and astrophysics through PINNs

Integration Across Themes: essential AI/ML tools for experimental analysis, closure model development, and next-generation hardware co-design for scientific computing

Feasibility: established leadership in structure-preserving neural networks, tensor methods, neuromorphic computing and multi-scale modeling

Legacy: Establish new paradigm for scientific AI/ML that preserves physical structure, enabling predictive modeling of complex multi-scale phenomena across disciplines

(3c) Education and Human Resource Development Objectives

TEMPEST's education program embodies the transformative potential of integrating cutting-edge research with innovative pedagogy to create the next generation of non-canonical turbulence researchers. Our approach recognizes that solving the complex challenges of non-canonical turbulence requires scientists who can seamlessly traverse experimental, computational, and theoretical domains — a capability that traditional disciplinary education cannot provide. Through comprehensive integration of research and education across all levels, from K-12 engagement to postdoctoral training, TEMPEST will create a globally competitive workforce equipped to tackle multi-scale, multi-physics problems that define the frontiers of modern science and engineering.

The foundation of our education strategy lies in breaking down the traditional barriers between experimental, computational, and theoretical approaches through an innovative "inreach" program that ensures all TEMPEST researchers can communicate across disciplinary boundaries. Graduate students, postdocs, and undergraduate researchers will participate in intensive cross-training sessions, where experimentalists learn the challenges and opportunities of computational modeling, while theorists gain hands-on experience with experimental constraints and data interpretation. This cross-fertilization occurs through structured immersion periods where students from different disciplines work together for weeks at a time, sharing facilities, datasets, and research challenges. All TEMPEST data, from individual particle tracking in dusty plasmas to exascale simulations, will be available to every student and postdoc, creating unprecedented opportunities for interdisciplinary discovery and learning.

Our graduate education program centers on developing researchers who embody the integrative spirit essential for boundary turbulence research. Every postdoctoral researcher will have dual mentors from different disciplinary backgrounds, ensuring exposure to multiple research methodologies and perspectives throughout their training. Graduate students will participate in novel courses that connect directly to TEMPEST's three research pillars, gaining experience with revolutionary experimental facilities, advanced AI/ML techniques for turbulence analysis, and multi-scale theoretical frameworks. Travel support enables students to work at partner institutions, national laboratories, and industry sites, creating a truly distributed educational experience that leverages the full breadth of TEMPEST's collaborative network. Summer internship programs with Los Alamos and Lawrence Livermore National Laboratories, and industry partners like Pacific Fusion, provide real-world research experiences that bridge fundamental science and practical applications.

Undergraduate research integration represents a cornerstone of TEMPEST's broader impact strategy, embedding students directly in center research activities while developing computational and scientific literacy for the broader student population. The interdisciplinary course "Scientific Computing Literacy in the Age of AI" engages undergraduates from all majors in understanding how models, simulations, and artificial intelligence inform modern science, using authentic TEMPEST datasets to explore phenomena spanning extreme weather to astrophysics. Students will work with real data from our seven types of non-canonical turbulence experiments, applying computational thinking in contexts that mirror authentic research [61] while contributing to ongoing research efforts through guided analysis and model validation. This approach builds computational confidence and scientific identity among diverse student populations, creating pathways for students from traditionally underfunded groups to engage with cutting-edge STEM research [32].

Our K-12 engagement strategy leverages TEMPEST's unique research capabilities to create authentic scientific experiences that inspire the next generation of researchers. Modular micro-courses designed for high school learners integrate simplified datasets from actual TEMPEST research, allowing students to engage with real scientific challenges through inquiry-driven activities. These modules prioritize accessibility and relevance, requiring no prior coding experience while still exposing students to computational practices and data interpretation. The design emphasizes hands-on, problem-based learning that connects non-canonical turbulence concepts to phenomena students encounter in daily life, from weather patterns to energy systems. Professional development workshops for educators provide comprehensive support for implementing these materials, including immersive summer institutes where teachers engage directly with TEMPEST research and develop confidence in facilitating computational science experiences.

The assessment and evaluation framework for TEMPEST's education program combines rigorous measurements of learning outcomes with continuous improvement using evidence-based practices. We will measure both cognitive gains in understanding turbulence concepts and attitudinal shifts toward STEM careers, with particular attention to students from non-STEM backgrounds and students from traditionally underfunded groups [62]. Custom assessment tools will evaluate students' grasp of multi-scale physics concepts, data interpretation skills, and computational thinking abilities, while validated surveys measure changes in STEM interest and self-efficacy. Log data from learning platforms will provide quantitative insights into student engagement and participation patterns. Results will directly inform iterative curriculum refinement, ensuring that educational materials remain effective for a wide range of learners while maintaining connection to cutting-edge research advances. This effort is in collaboration with our external assessors at Strategic Evaluation, Inc. (SEI), which is discussed in the management section.

TEMPEST's education program creates sustainable impact through its integration with the center's research mission and external partnerships. Students at all levels contribute to actual research discoveries while developing skills essential for modern scientific careers. The distributed nature of our education activities, spanning multiple institutions and leveraging national laboratory and industry partnerships, creates a model for collaborative education that extends far beyond traditional academic boundaries. By training researchers who can work seamlessly across experimental, computational, and theoretical domains, TEMPEST addresses a critical workforce need in fields from basic fusion science to aerospace engineering. The program's emphasis on authentic research experiences and cross-disciplinary collaboration prepares graduates to tackle the complex, multi-scale challenges that define the frontiers of 21st-century science and technology.

Broader Impacts

(3d) Broadening Participation Objectives of the Center

TEMPEST will transform turbulence science by promoting it at all levels—from K-12 and public engagement through early-career training—to build a broadly based, resilient research community that reflects the full spectrum of human talent and perspective.

Our coordinated turbulence science outreach will reach 10,000+ students annually through experienced museum and public engagement leaders: Aluie (University of Rochester with the Rochester Museum and Science Center), Matthews (Baylor University with the Mayborn Museum), Yuan/Huang (MSU), Li (TAMU-CC), and Kostadinova (Auburn). We will deploy our touring centerpiece art exhibition to transform high-traffic halls in partner institutions with immersive ceiling/wall projections where visitors physically manipulate multi-scale turbulent flows, making complex physics tangible and approachable.

Global impacts will include a YouTube channel building scientific literacy leveraging PI Murillo's success (750,000+ TED-Ed views), the "Turbulent World" art competition connecting creativity to multi-scale turbulence science, and immersive technologies combining VR plasma turbulence visualization (leveraging Kostadinova's APS PhysicsQuest 2024 experience) with a multiplayer galaxy creation browser game via MSU's top-10 Games for Entertainment and Learning Lab (creators of 5.0-rated Quantum 3 mobile game on quantum chromodynamics), plus an Unreal Engine plug-in bringing advanced multi-physics modeling to gaming/film industries. At the undergraduate level, research internships and the interdisciplinary course "Scientific Computing Literacy in the Age of AI" will build computational confidence and scientific identity across all backgrounds, with intentional outreach to under-resourced institutions and economically disadvantaged communities.

Graduate training will revolutionize traditional approaches through 1-month cross-institutional student exchange programs, biannual 5-day collaboration retreats (leveraging Collaborate@ICERM expertise), and an annual 4-day TEMPEST Turbulence Symposium that immerse trainees across experimental, computational, and theoretical environments, with formal cross-institutional mentoring connecting each trainee to distributed teams. TEMPEST integration was already proven effective during proposal development: cross-disciplinary concept exchange sessions revealed disciplinary language obstacles that masked shared fundamental concepts (e.g., terms such as "turbulence" and "correlation" with different surface meanings but coherent underlying physics). Professional development will center on annual communication-focused summer schools building on proven AIDMM NRT expertise and external partnerships (e.g., Science Trust and APS Public Engagement Office), leveraging Promislow's and Kostadinova's experiences. The curriculum will develop all-audience science communication and proposal writing skills essential for modern careers.

The MSU Turbulence Library will operate as an open science hub where all trainees contribute datasets, codes, and educational content as thesis requirements, supported by FAIR (findable, accessible, interoperable, and reusable) Data and open science training. This approach will inculcate data stewardship skills while make cutting-edge research output more available. These integrated efforts will accelerate innovation and workforce transformation by creating a new interdisciplinary scientist archetype capable of seamlessly navigating experimental, computational, and theoretical domains—addressing complex multi-scale scientific challenges through the full spectrum of human talent and perspective.

TEMPEST employs an external evaluation firm to monitor broadening participation through demographic tracking, institutional climate assessments, regular stakeholder feedback, and integrated management reviews that inform center practices. See details in Sec. (3f).

(3e) Knowledge Transfer Objectives of the Center

Knowledge transfer lies at the heart of the TEMPEST mission to transform fundamental advances in non-canonical turbulence into practical solutions for the most pressing challenges in society. Our approach emphasizes an exchange of scientific and technical information with external stakeholders, ensuring that our research both addresses real-world needs and benefits from the expertise and perspectives of industry, national laboratories, and policy-making organizations. This strategy builds on the established networks and collaborative relationships that our team members have cultivated throughout their careers, creating pathways for immediate impact while establishing sustainable mechanisms for long-term knowledge exchange.

Our knowledge transfer strategy centers on three primary categories of external stakeholders, each representing critical applications of non-canonical turbulence research. National laboratories form our most established partnership network, with particularly strong connections to those in the DOE. These partnerships leverage existing relationships built through former students now working at these institutions, as well as ongoing collaborative research projects and formal appointments. Industry partnerships focus on organizations at the forefront of fusion energy and plasma technology, including Zap Energy, Pacific Fusion, and General Atomics, where our advances in plasma turbulence modeling can directly accelerate the development of commercial fusion energy systems. Such policy and regulatory stakeholders (e.g., DOE, NOAA, and NASA) will benefit from improved turbulence model to enhance everything from fusion energy program planning to climate prediction capabilities and space weather forecasting.

The specific knowledge products that TEMPEST will transfer span the full spectrum from fundamental datasets to practical software tools. Our experimental facilities will generate unprecedented datasets for many types of non-canonical turbulence, including 1) particle-resolved dusty plasma flows, 2) shock-boundary layer interactions, 3) variable viscosity flows, 4) fluid-structure interactions, 5) volumetric energy deposition instabilities, 6) compressible turbulence at extreme conditions, and 7) multi-scale mixing processes. These datasets will provide the validation benchmarks that external partners need to improve their own modeling capabilities. We will develop and release advanced software tools and algorithms, with particular emphasis on improving existing computational codes such as FLASH. Our primary focus on developing improved turbulence closures addresses a critical need across multiple application domains.

Our knowledge transfer mechanisms emphasize sustained, meaningful intellectual exchange, as student internship programs will place TEMPEST researchers in national laboratory and industry settings for extended periods. The partnerships that already exist with in the TEMPEST team demonstrate proven pathways for knowledge transfer, with facilities like the VED already serving as platforms for training students from diverse backgrounds.

The cyber-infrastructure component of our knowledge transfer strategy leverages modern data sharing and collaboration technologies to extend our impact far beyond our direct partnership network. All experimental and simulation datasets will be made available through open repositories with comprehensive metadata and standardized formats. We will implement tracking mechanisms to monitor dataset downloads and usage, providing quantitative measures of our knowledge transfer impact. Advanced visualization and analysis tools will accompany these datasets, simplifying the process for external users to extract value from our research, and virtual collaboration platforms will enable real-time interaction between TEMPEST researchers and external partners.

Our approach to monitoring knowledge transfer impact combines quantitative metrics with qualitative assessment of societal benefit, where we will monitor the practical implementation of our advances through regular communication with our industry and national laboratory partners. Success stories will be documented and shared to demonstrate the value of fundamental non-canonical turbulence research to both the scientific community and funding agencies.

TEMPEST's knowledge transfer strategy recognizes that the most transformative advances emerge from sustained collaboration between fundamental researchers and practitioners facing real-world challenges. By embedding knowledge transfer activities throughout our research program rather than treating them as an afterthought, we ensure that our scientific discoveries rapidly translate into societal benefits. This approach continues the tradition of turbulence research as a field that combines intellectual curiosity with tangible impact, positioning TEMPEST to deliver both scientific breakthroughs and practical solutions that address humanity's most pressing challenges.

TEMPEST's comprehensive research strategy culminates in concrete, measurable outcomes that demonstrate the center's transformative impact across all mission areas. Table 1 summarizes **key milestones for Years 2 and 5**, illustrating how our integrated approach progresses from foundational development in experimental capabilities, theoretical frameworks, and AI/ML methodologies to full-scale demonstration of breakthrough technologies and their translation into practical applications. These milestones reflect our commitment to knowledge transfer as a core principle, ensuring that advances in each research pillar directly contribute to solving grand challenges in fusion energy, climate prediction, and engineering systems while building the next generation of interdisciplinary turbulence scientists.

Table 1: TEMPEST Center key milestones across five integrated research pillars for Years 2 and 5. The center's approach emphasizes knowledge transfer as a core component embedded within each research area, ensuring that advances in experimental capabilities, theoretical frameworks, AI/ML methodologies, open science practices, and educational programs translate directly into practical applications for fusion energy, climate modeling, and engineering systems.

Year 2	**Year 5**
Experimental	
Datasets: dusty plasma kinetic-to-fluid scales, shock-turbulence interactions, variable viscosity, energy-density-momentum coupling; *Framework:* HED materials transport coefficient.	*Datasets:* multi-physics. *Analysis:* non-Maxwellian distributions, molecular chaos assumption breakdown documented. *Validations:* XPCS transport coefficient database, cross-scale energy transfer.
Theory & Modeling	
Development: fractional calculus models, nonlocal theories, multiscale coarse-graining frameworks, BBGKY hierarchy closures, open-source FlowSieve extended.	*Demonstration:* multi-physics closures bridging particle-to-collective scales, structure-preserving methods, models for experimental facilities, and exascale computing integration.
AI/ML & Data Science	
Development: structure-preserving neural networks, physics-informed ML frameworks, automated closure discovery tools, cross-facility data fusion algorithms, tensor decomposition for multi-physics, sparse turbulence analysis.	*Demonstration:* AI hybrid models, atomic to turbulence scales, neuromorphic real-time predictions, ML extreme event prediction framework, automated scientific discovery pipeline, advanced UQ, adaptive sampling optimization.
Open Science & Data	
Launch: TEMPEST Turbulence Library, API access for real-time sharing, repos for all analysis codes, DOIs for datasets, open-source tools released.	*Accomplished:* 100TB+ open data, validated DNS/MD databases, open-access publications/codes, interactive visualization, machine-readable metadata standards.
Education & Impact	
Deploy: cross-disciplinary rotation program, professional development/data literacy training, K-12 micro-courses, interactive and digital outreach, industry/national lab internships.	*Deliver:* 100+ multi-physics trained students, national educator network, computational curricula for 50+ schools, art/gaming public engagement, technology transfer to commercial codes.

(3f) Management Plan

TEMPEST's management structure combines proven leadership with innovative coordination mechanisms across seven partner institutions. Our distributed, adaptive model balances centralized strategic direction with agile execution, enabling rapid response to scientific opportunities while maintaining accountability.

Governance Structure. Michigan State University leads a coalition including Auburn, Baylor, Georgia Tech, Rochester, Texas A&M Corpus Christi, Yale, three national laboratories (Los Alamos, Lawrence Livermore, Sandia), and industry partners (General Atomics, Pacific Fusion, Zap Energy). The External Advisory Board provides independent strategic guidance to Center

Director Prof. Michael Murillo, supported by Project Manager, Communications Specialist, and Administrative Assistant.

Prof. Murillo brings extensive multi-institutional leadership experience as a former Los Alamos Project/Team Leader, current IEEE ICOPS 2026 General Chair, Physical Review E Editorial Board member, and APS Fellow. His broadening participation commitment includes widely viewed TED-Ed content (750,000+ views) and teaching excellence (2024 Fritz Teaching Excellence Award). He develops DS and ML courses for scientists and engineers.

Five Co-Directors provide specialized leadership: Research Co-Director Prof. Anna Gilbert (Yale) coordinates AI/ML pillar and manages integration across scales; Research Co-Director Prof. Qi Tang (Georgia Tech) coordinates theory/modeling pillar and facilitates integration; Research Co-Director Prof. Evdokiya Kostadinova (Auburn) with co-lead Prof. Ricardo Mejia-Alvarez (MSU) oversees experimental integration; Education/Broadening Participation Co-Director Prof. Junlin Yuan (MSU) with Prof. Aman Yadav (MSU) leads education and outreach; Knowledge Transfer Co-Director Prof. Andrew Christlieb (MSU) manages industry partnerships/technology transfer.

The Executive Committee coordinates strategic initiatives across mission areas. Major Activity Leaders for AI/ML, Theory & Modeling, and Experiments serve rotating one-year terms, enabling domain expert leadership while developing early-career researchers through structured leadership shadowing programs. Research Area Leaders co-lead five areas (Scale Interactions, Structure Formation, Transport Mechanisms, Model Validation, Cross-Cutting Technologies) across institutions. A Student and Postdoc Board ensures early-career representation and supports reverse mentoring.

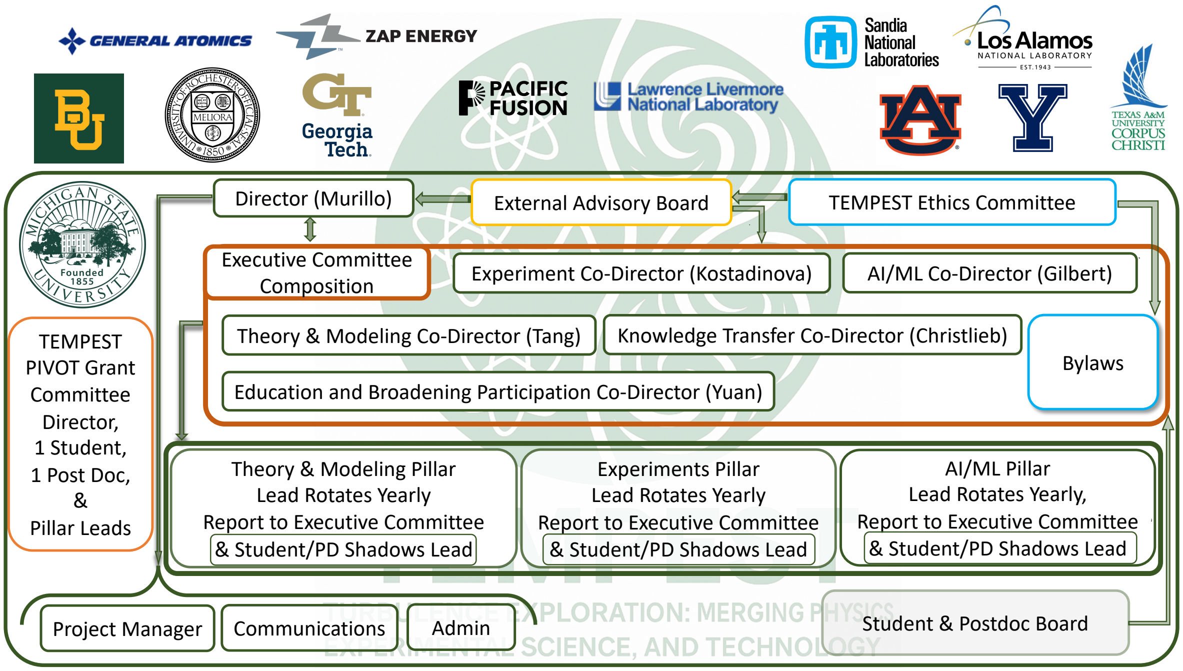

Figure 4: TEMPEST organizational structure showing partner institutions (top), External Advisory Board and Director oversight, Executive Committee with Co-Directors managing bylaws, three research pillars with rotating leads, PIVOT internal grant program, and administrative support with Student/Postdoc Board representation.

Integration and Coordination. TEMPEST's collaboration model centers on one-month cross-disciplinary rotation programs embedding students and postdocs outside their primary areas. AI/ML researchers work with experimental/theory groups and vice versa, building a common scientific language and training researchers capable of navigating multi-scale turbulent system complexity. Institutional Ambassadors spend extended time at partner institutions, deepening collaborative networks and identifying synergies. Research coordination occurs through monthly thrust group meetings, center-wide meetings with rotating scientific talks, and PI lecture series. Interdisciplinary innovation sprints prototype new ideas, while failure retrospectives foster creativity without penalty. Major facilities (ABC, MPRL, dusty plasma chambers) operate under shared governance with standardized protocols. Digital collaboration employs integrated tools (Slack, Notion, shared repositories) with regular videoconferences and shared computational resources. Full-time Communications Coordinator manages external visibility and internal messaging, working with Project Manager and Administrative Assistant to maintain coordination.

Strategic Planning and Resource Allocation. Annual strategic reviews assess progress, identify opportunities, and adjust allocation based on research area input, advisory recommendations, and stakeholder feedback. Projects are prioritized by alignment with our non-canonical turbulence mission, breakthrough potential, integration opportunities, and broader impact. The Executive Committee maintains governing bylaws with full community input; the External Advisory Board arbitrates disputes. Resources are allocated transparently: core support based on deliverables/milestones, competitive internal funding for high-risk initiatives, infrastructure maintenance, and integration activities. The Director makes final decisions based on advisory input.

PIVOT Program. TEMPEST's Internal Grant Program PIVOT (Proactive Innovation Vision Opportunities in Turbulence) revolutionizes center operations by transforming static research plans into a dynamic, adaptive ecosystem that captures breakthrough opportunities as they emerge. PIVOT addresses NSF's vision for centers that exploit complexity, requiring unprecedented scope, scale, flexibility, and duration by creating institutional agility impossible in traditional research structures. Five strategic tiers enable rapid scientific pivoting: Rapid Response Grants capture time-sensitive opportunities with accelerated review; Inreach and Integration Grants strengthen cross-institutional collaboration and method transfer; External Integration Grants bring world-class external expertise into the TEMPEST ecosystem; Innovation Grants enable bold strategic redirections toward high-impact discoveries; and Broadening Participation Grants expand access to under-resourced communities, for example through an award to scale TARDIS Summer of Code computational internships for community college students, leveraging TEMPEST member Kerzendorf's established partnerships with 25 Michigan community colleges. A tiered committee structure ensures rigorous evaluation and strategic alignment: small grants (<\$50K) reviewed by pillar leads, medium grants (<\$150K) evaluated by the Executive Committee, and transformative grants (>\$150K) assessed by a full Center Committee including external advisors, all operating in consultation with the Director to rank and prioritize new activities against TEMPEST's evolving strategic objectives. This creates a scientific venture capital model within the center—teams propose high-risk/high-reward concepts for seed funding, successful initiatives receive enhanced support and integration, while the committee structure ensures alignment with center goals and maximizes scientific impact. PIVOT transforms TEMPEST from a coordinated research program into an adaptive discovery engine capable of responding to breakthrough opportunities faster than any traditional funding mechanism.

Lead Institution Infrastructure and Personnel Commitments. Michigan State University demonstrates substantial institutional commitment to TEMPEST through strategic space planning that reflects the center's interdisciplinary scope and anticipated growth. *Center Operations*: MSU has committed approximately 2,100 ft^2 of dedicated space for core TEMPEST operations, including director and administrative offices, visitor accommodations, collaborative work areas, and multimedia facilities, centrally located to serve as the primary hub for center coordination and external partnerships. *Research Infrastructure*: Building on existing experimental capabilities across the Engineering building and Engineering Research Center, MSU will accommodate additional laboratory space for new experimental facilities essential to TEMPEST's mission, with flexible space management enabling optimal equipment placement based on technical requirements and collaborative access needs. *Faculty Development*: MSU's strategic investment in turbulence and AI research aligns directly with TEMPEST objectives. The university is well-positioned to pursue key faculty hires supported by MSU's institution-wide AI initiative and the College of Engineering's focus on expanding computational and experimental turbulence capabilities. TEMPEST funding will significantly accelerate these hiring priorities, enabling recruitment of top-tier talent in turbulence-AI intersections. *Cross-College Integration*: The partnership between Engineering and Natural Science provides comprehensive institutional support, with both colleges prepared to contribute resources supporting TEMPEST's interdisciplinary approach and educational mission.

Performance Monitoring. Strategic Evaluations, Inc. (SEI), a HUB-certified minority-owned firm, provides independent evaluation. Director Dawayne Whittington's team brings 15+ years experience with multi-institutional STEM centers, currently evaluating Ohio State NRT. SEI employs mixed-methods (surveys, interviews, analytics) with quarterly formative and annual summative assessments informing real-time adjustments and strategic planning. Our comprehensive theory of action model connects center inputs and activities to measurable outputs and long-term impacts across five domains. Research metrics track interdisciplinary publications, co-authorship networks, tool uptake metrics, and partner citations, with validated 10^{12}-scale turbulence models assessed through expert review panels. Education effectiveness indicators include course enrollment/completion rates, learning assessments, REU tracking, student career interest surveys, and alumni career trajectories across all educational levels from K-12 through graduate training. Broadening participation metrics monitor demographic participation data, pre/post engagement surveys, community stakeholder interviews, and partnership effectiveness with resource-limited institutions and community colleges through structured assessment tools including the Partnership Self-Assessment Tool measuring leadership, efficiency, administration, and resource sharing. Knowledge transfer indicators encompass software/open data download statistics, feedback from labs and industry partners, partnership reports, and technology adoption challenges identified through structured partner interviews. Center operations are evaluated through meeting logs, network analysis, stakeholder surveys, and Advisory Board reports assessing communication effectiveness, collaboration quality, and progress toward sustainability beyond NSF funding. We utilize evidence-based tools including scientific integration measures (self-efficacy, identity, shared values) adapted from validated frameworks to document trainee growth and persistence in STEM careers. Quarterly risk assessments address coordination, engagement, and allocation challenges, with evaluation findings directly informing quarterly leadership reviews and annual strategic planning sessions. An initial, detailed *Logic Model* has been developed for TEMPEST (not shown).

Risk Management and Succession. TEMPEST employs proactive risk mitigation across multiple dimensions: coordination risks are addressed through structured communication protocols, quarterly cross-pillar integration meetings, and adaptive management mechanisms; technical risks are mitigated through diversified experimental platforms, redundant computational approaches, and validation across multiple partner institutions; personnel risks are managed through distributed leadership, cross-training of key personnel, and substantial partner contributions in research time, laboratory access, and specialized equipment. Financial sustainability is ensured through diversified funding portfolios and demonstrated multi-institutional coordination capabilities. Formal succession planning identifies and develops future directors among Co-Directors and senior faculty through progressive leadership responsibilities, advisory board interaction, NSF reporting experience, and structured mentoring programs that build next-generation center leadership capacity. These comprehensive risk mitigation strategies ensure TEMPEST's long-term sustainability and continued leadership in non-canonical turbulence research.

External Advisory Committee. Representatives from industry, national laboratories, government, and academia provide independent oversight and strategic guidance. Annual evaluation of training effectiveness, management advice, and policy recommendations ensure objective assessment of center progress toward NSF goals, with dispute arbitration and succession guidance.

(3g) Institutional Commitment to Promoting Inclusion

Michigan State University and its partner institutions demonstrate substantial commitment to broadening participation in STEM through strategic institutional programs. MSU's recent NSF ADVANCE grant (2305598) strengthens institutional transformation for academic workforce development, while the internally funded 1855 Professorships Initiative has created three faculty positions in CMSE focused on increasing representation. The university's Sloan Foundation partnerships with institutions including the Atlantic University Consortium create pathways to foster representation in STEM careers, supported by MSU's administration through public engagement via the MSU Science Festival, Museum, and Science Center. Partner institutions contribute complementary strengths: San José State University extends impact through specialized STEM programs across the California State University system, while partnerships with Baylor University and Los Alamos National Laboratory provide unique opportunities in plasma research for students from comprehensive educational backgrounds, collectively creating an institutional ecosystem that supports TEMPEST's mission to expand accessibility and strengthen connections with regional institutions, community colleges, and K-12 schools.

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